② 1. $\log p(x|0) = \log \frac{p(0|x)\,p(x)}{p(0)} = \log p(0|x) + \log p(x) + C_1 =$

$= -\frac{1}{2}\left((0-Hx)^T R^{-1}(0-Hx)\right) + -\frac{1}{2}\left((x-m_0)^T \Sigma_0^{-1}(x-m_0)\right) +$

$+ C_2 = -\frac{1}{2}(x-\bar{x})^T \bar{\Sigma}^{-1}(x-x^T) + C_3$

Розкриємо дужки і прирівняємо:

- $-20^T R^{-1} \cdot Hx + x^T H^T R^{-1} Hx + x^T \Sigma_0^{-1} x - 2m_0^T \Sigma_0^{-1} x =$
  $= x^T(H^T R^{-1} H + \Sigma_0^{-1}) - 2(0^T R^{-1} H + m_0^T \Sigma_0^{-1})x$
- $x \bar{\Sigma}^{-1} x - 2\bar{x}^T \bar{\Sigma}^{-1} x$

$\therefore$ $H^T R^{-1} H + \Sigma_0^{-1} = \bar{\Sigma}^{-1}$, $0^T R^{-1} H + m_0^T \Sigma_0^{-1} = \bar{x}^T \bar{\Sigma}^{-1}$.

$\bar{\Sigma} = (H^T R^{-1} H + \Sigma_0^{-1})^{-1}$

$\bar{x} = \bar{\Sigma}\left(H^T R^{-1} 0 + \Sigma_0^{-1} m_0\right) =$

$= (H^T R^{-1} H + \Sigma_0^{-1})^{-1}\left(H^T R^{-1} 0 + \Sigma_0^{-1} m_0\right)$

Put $m_0 = 0$, $\hat{x}_{MAP} = \left(\Sigma_0^{-1} + H^T R^{-1} H\right)^{-1} H^T R^{-1} 0$ — 100% match

2. $\hat{x}_{MAP} = \underset{D=\{x=(x_1,x_2):\, x_1+x_2 \le 2\}}{\operatorname{argmin}} \|0 - x\|^2 + x^T \Sigma_0^{-1} x =$

$$= \underset{x \in D}{\operatorname{argmin}} \left\{ (x_1 - 3)^2 + (x_2 - 1)^2 + \frac{1}{4} x_1^2 + x_2^2 \right\} =$$

$$= \underset{x \in D}{\operatorname{argmin}} \left\{ 5x_1^2 + 8x_2^2 - 24x_1 - 8x_2 \right\}$$

Маємо такий Лагранжіан:

$$\mathcal{L}(x, \mu) = 5x_1^2 + 8x_2^2 - 24x_1 - 8x_2 + \mu(x_1 + x_2 - 2)$$

$$\nabla \mathcal{L} = \begin{pmatrix} 10x_1 - 24 + \mu \\ 16x_2 - 8 + \mu \\ x_1 + x_2 - 2 \end{pmatrix} = 0 \iff \begin{cases} 8x_2 - 5x_1 + 8 = 0 \\ x_1 + x_2 - 2 = 0 \end{cases} \quad 13x_2 = 2 \quad \begin{cases} x_2 = \frac{2}{13} \\ x_1 = \frac{24}{13} \end{cases}$$

$$\mu = 8 - 16x_2 = 8 - \frac{32}{13} = \frac{72}{13} > 0 \ \checkmark$$

$$\begin{pmatrix} x_2 \\ x_1 \end{pmatrix} = \begin{pmatrix} 2/13 \\ 24/13 \end{pmatrix} - \text{active, because } x_1 + x_2 = 2.$$

Let's find unconstrained x_{MAP}:

$$x_{MAP} = \left(\Sigma_0^{-1} + I\right)^{-1} 0 = \operatorname{diag}\left(\frac{5}{4}, 1\right)^{-1} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 12/5 \\ 1 \end{pmatrix}$$

$$\Pr_{x_1 + x_2 = 2}\left(\begin{pmatrix} 12/5 \\ 1 \end{pmatrix}\right) = \left(x_{MAP} - \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}\right) \begin{pmatrix} -1 \\ 1 \end{pmatrix} \Big/ 2 + \begin{pmatrix} 1 \\ 1 \end{pmatrix} =$$

$$= \begin{pmatrix} -\frac{1}{2} \\ 1 \end{pmatrix} \left(\left(\frac{7}{5}, 0 \right), \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right) / 2 + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = -\frac{7}{10} \cdot \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 17/10 \\ 3/10 \end{pmatrix}.$$

0 - ближче до $\hat{x}_{MAP}$ ніж x_{MAP}



